Tritium breeding for fusion power and the LIBRA project

MIT PSFC

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Three main ideas for this presentation.

1. Efficient tritium breeding is required for a successful fusion power industry.

2. Liquid immersion blanket (LIB) is a competitive tritium breeding concept.

3. The PSFC will use the LIBRA experiment to develop LIB technology.

The basics of tritium (T) breeding for D-T fusion

D + T → α + n + heat

D-T fueled power plants eventually need to **breed** T fuel.

D-T fusion reactions in the plasma generate **14 MeV neutrons**. We use these to generating heat and T in a breeder blanket.

TBR = tritium breeding ratio.
If TBR = 1, one triton is produced in the blanket for every triton consumed by D-T reactions.

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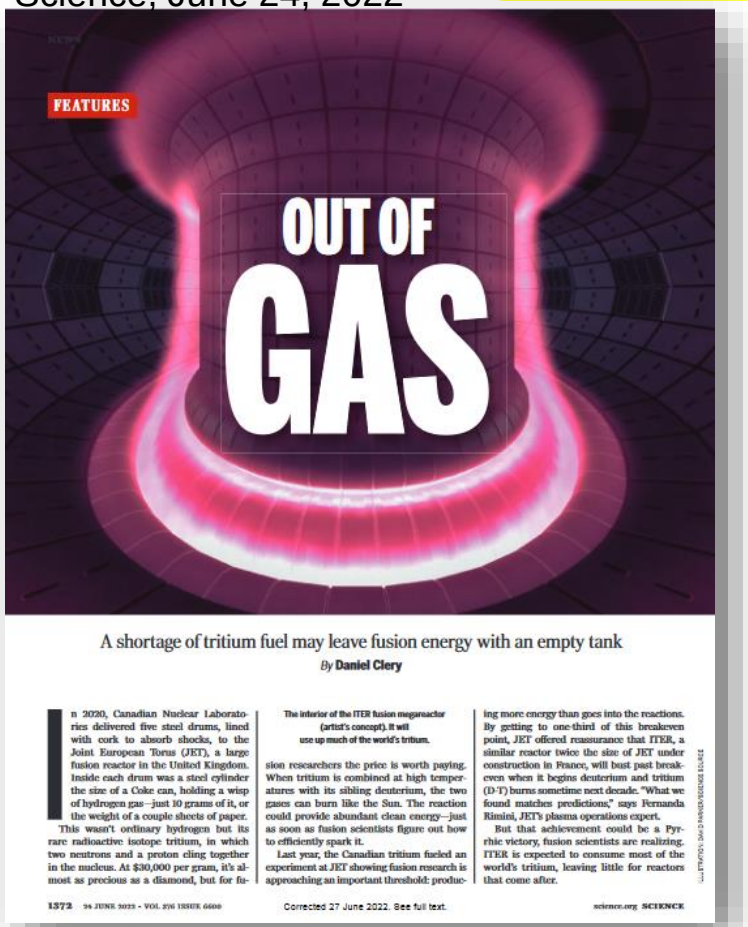
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~56 kg of T consumed/1 GW fusion power/year
<0.5 kg of T/year produced in fission reactors specifically designed for T production

Projected 28 kg peak T inventory before year 2030

Practical tritium supply comes from fission reactors

Science, June 24, 2022



FEATURES

OUT OF GAS

A shortage of tritium fuel may leave fusion energy with an empty tank

By Daniel Clery

In 2020, Canadian Nuclear Laboratories delivered five steel drums, lined with cork to absorb shocks, to the Joint European Torus (JET), a large fusion reactor in the United Kingdom. Inside each drum was a steel cylinder the size of a Coke can, holding a wisp of hydrogen gas—just 10 grams of it, or the weight of a couple sheets of paper.

This wasn't ordinary hydrogen but its rare radioactive isotope tritium, in which two neutrons and a proton cling together in the nucleus. At $30,000 per gram, it's almost as precious as a diamond, but for fusion researchers the price is worth paying. When tritium is combined at high temperatures with its sibling deuterium, the two gases can burn like the Sun. The reaction could provide abundant clean energy—just as soon as fusion scientists figure out how to efficiently spark it.

Last year, the Canadian tritium fueled an experiment at JET showing fusion research is approaching an important threshold: producing more energy than goes into the reactions. By getting to one-third of this breakeven point, JET offered reassurance that ITER, a similar reactor twice the size of JET under construction in France, will bust past breakeven when it begins deuterium and tritium (D-T) burns sometime next decade. "What we found matches predictions," says Fernanda Rimini, JET's plasma operations expert.

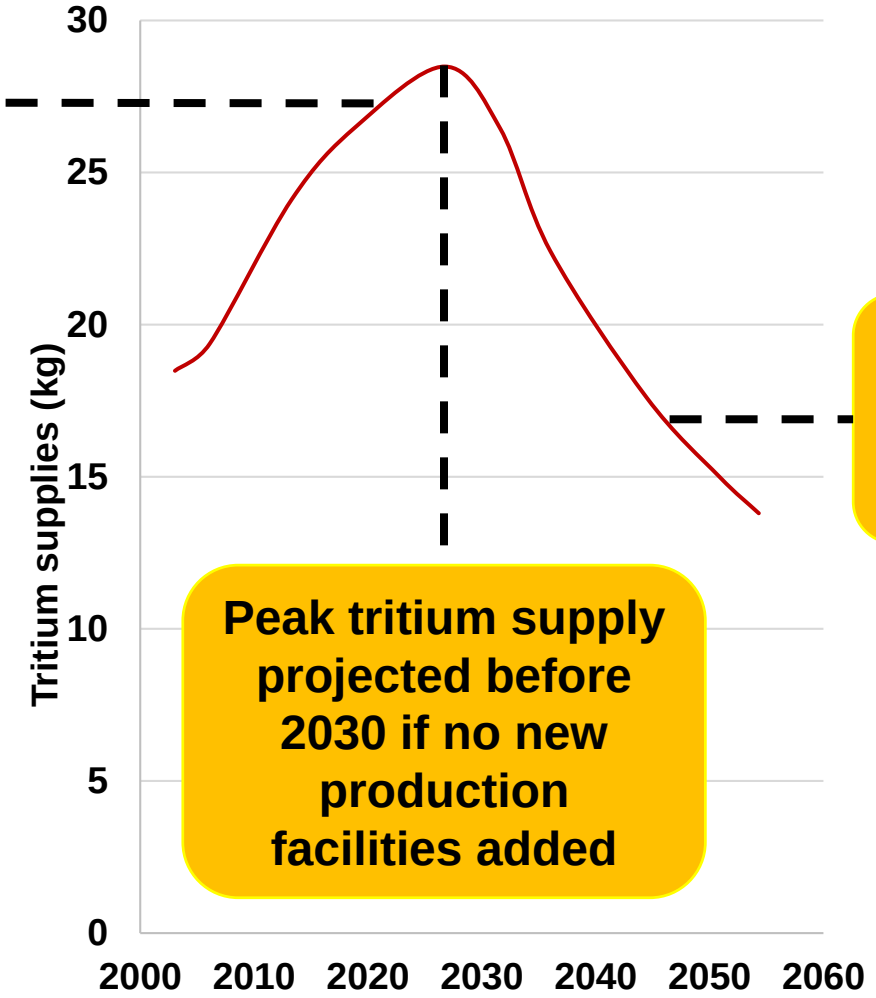
But that achievement could be a Pyrrhic victory, fusion scientists are realizing. ITER is expected to consume most of the world's tritium, leaving little for reactors that come after.

The interior of the ITER fusion megareactor (artist's concept). It will use up much of the world's tritium.

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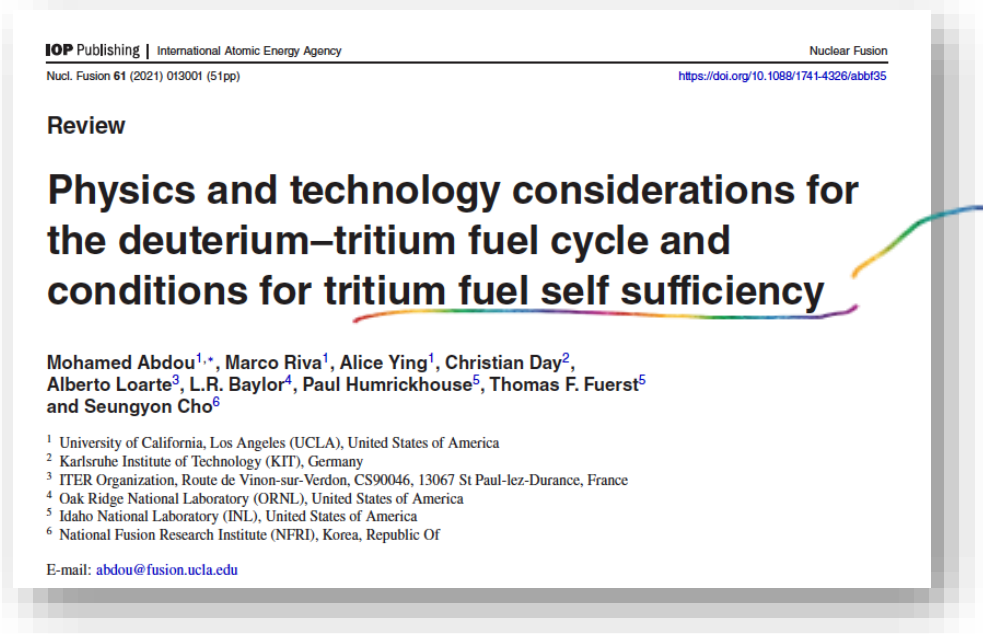
Tritium naturally decays away over time

Peak tritium supply projected before 2030 if no new production facilities added

World energy consumption in 2019 = 13.25 TW
World T supply in 2028 = 0.5 GW

Fuel self-sufficiency is required

Consider practical issues with tritium fuel cycles



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Nucl. Fusion 61 (2021) 013001 (51pp) https://doi.org/10.1088/1741-4326/abbf35

Review

Physics and technology considerations for the deuterium–tritium fuel cycle and conditions for tritium fuel self sufficiency

Mohamed Abdou[1,*], Marco Riva[1], Alice Ying[1], Christian Day[2], Alberto Loarte[3], L.R. Baylor[4], Paul Humrickhouse[5], Thomas F. Fuerst[5] and Seungyon Cho[6]

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[5] Idaho National Laboratory (INL), United States of America
[6] National Fusion Research Institute (NFRI), Korea, Republic Of

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$$\mathbf{TBR_{achievable} > TBR_{required}}$$

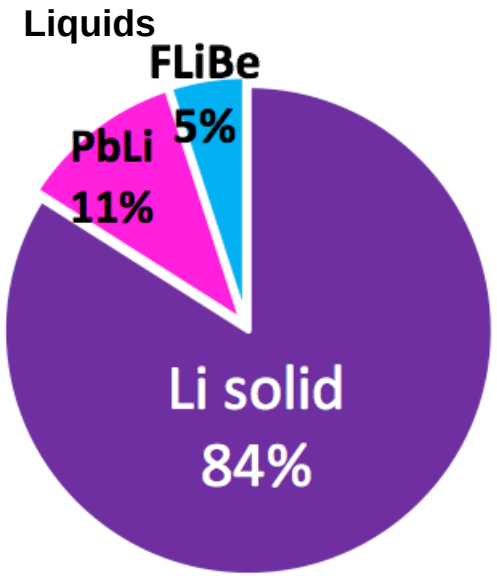
where $TBR_{required}$ accounts for:

- Self-fueling ← **TBR = 1**
- Compensate for decay losses
- Inventory for start-up of new plants
- Have some specified reserve inventory

TBR > 1

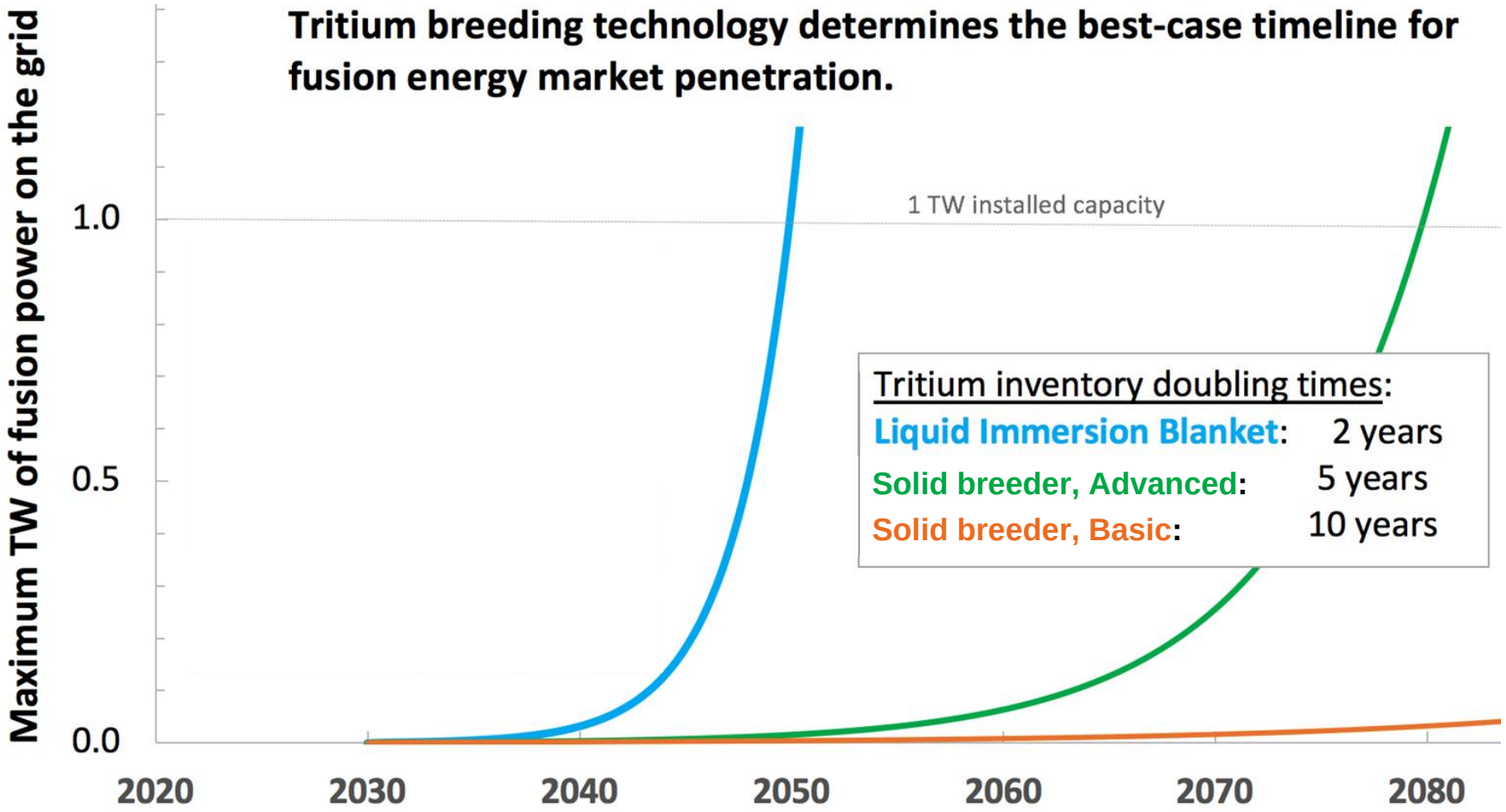
T breeding research

Experimental T breeding papers from last 50 years by breeder material category



- **$TBR_{achievable,\ solids} < TBR_{achievable,\ liquids}$**
- **Higher TBR allows faster growth of fusion industry**

High TBR and extraction efficiency has major implications for a fusion industry.



What need is there for T fuel without fusion power plants?

Increased funding and investment in fusion ~$2B over the past two years!

Rapid development to show net *scientific* power gain, Q>1 for fusion power out over heating power into core

U.S. "Bold Decadal Vision for Commercial Fusion Energy" White House Summit held in March 2022, U.K. government developing national program for net electricity

Commonwealth Fusion Systems (CFS) in Massachusetts leading the high-magnetic field approach with Q>1 demonstration planned for 2025-2026

Net fusion power gain in a few years proposed by CFS in MA

Planned CFS net fusion power gain test site in Devens, MA



cfs.energy

September 2022, tokamak hall



cfs.energy instagram

April 2022, magnet factory building



cfsatdevens.info

- **Tritium breeding blanket development for follow-on test for net electricity**

T breeding technology development in parallel to fusion core demonstration.

Key issues:

- **Tritium breeding potential – Conceptualized, small-scale tests**
- **Material compatibility – Corrosion, tritium transport, tritium speciation**
- **Material handling**
- **Tritium extraction methods**
- **Heat transfer**

Development ladder for tritium breeding technology

Scale difference in tritium production rate (TPR) and TBR:

Current tritium breeding tests	Fusion Power Plant
TBR<<1	TBR>1.15
TPR<10^{10} T/s	TPR>10^{20} T/s

Strategy:

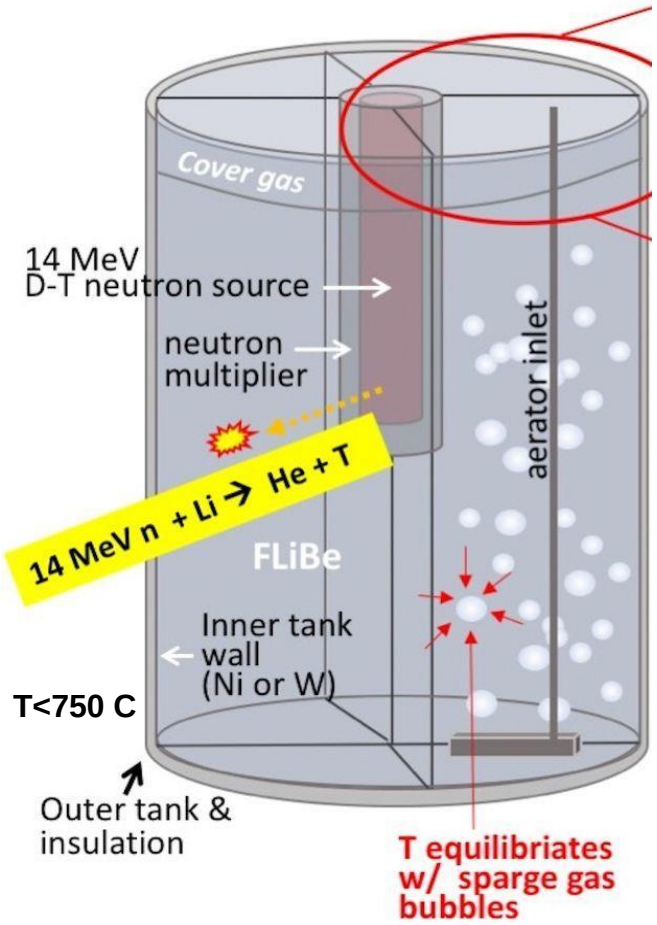
1. **Verify TBR up to 1 and validate concept.**
2. **Cover 10 orders of magnitude in tritium production rate in smaller steps**

LIBRA: Liquid Immersion Blanket Robust Accountancy

Breed T in FLiBe with D-T neutrons

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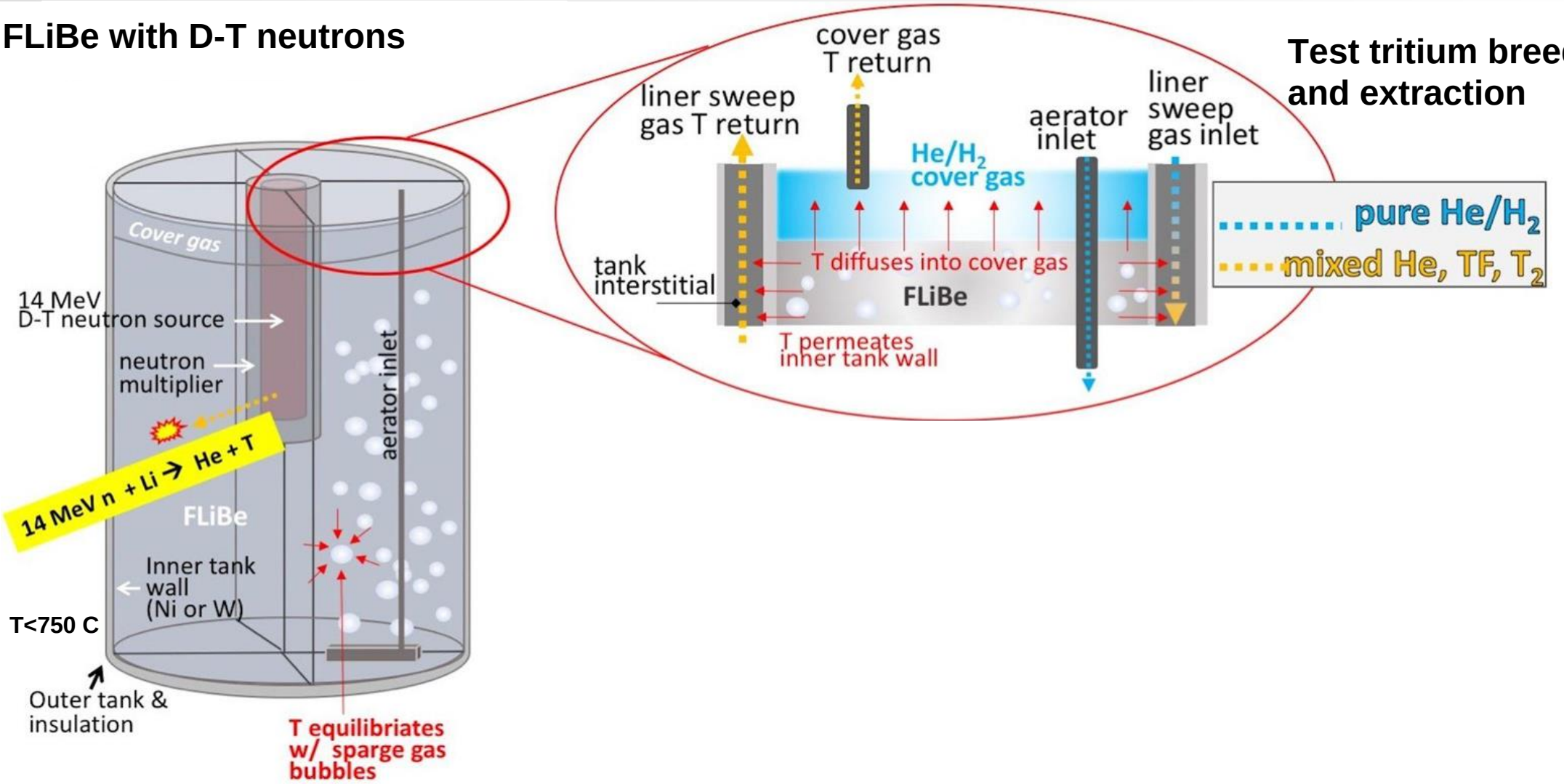


Modular tanks for parametric studies of materials, built-in repeatability, and future expansion

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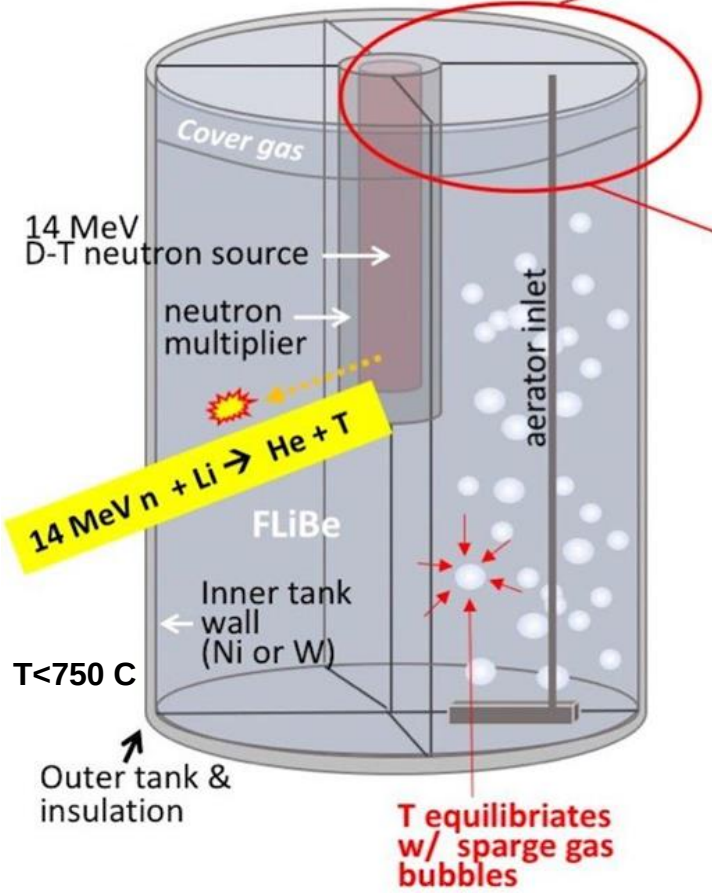
Test tritium breeding and extraction



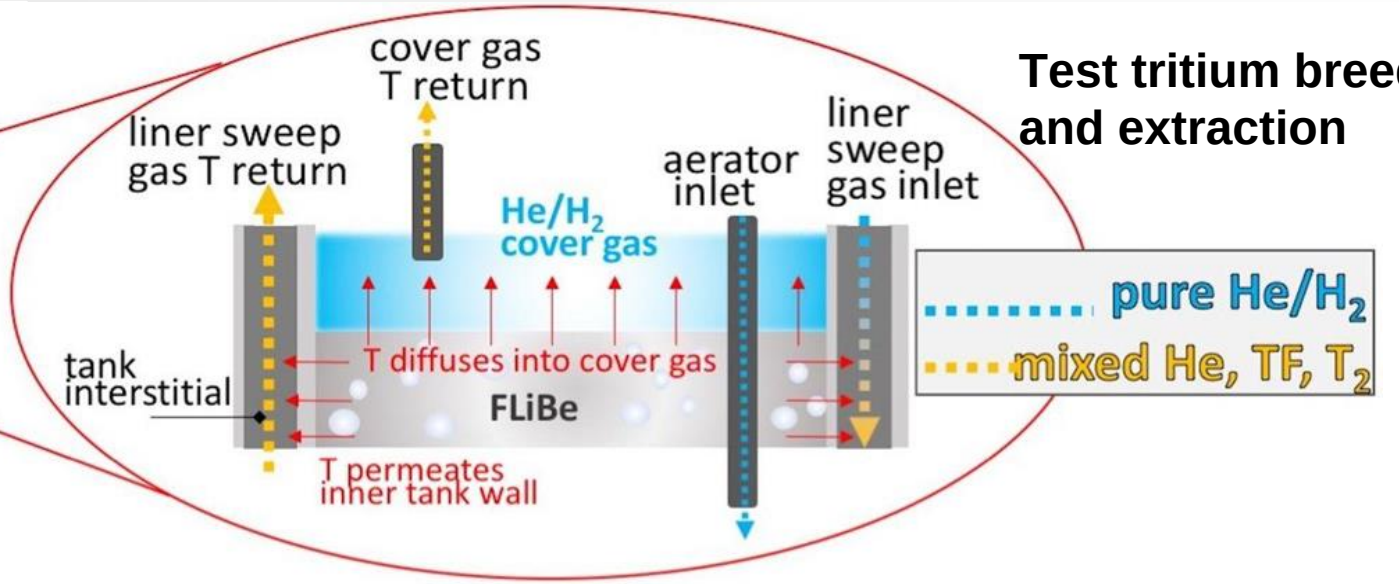
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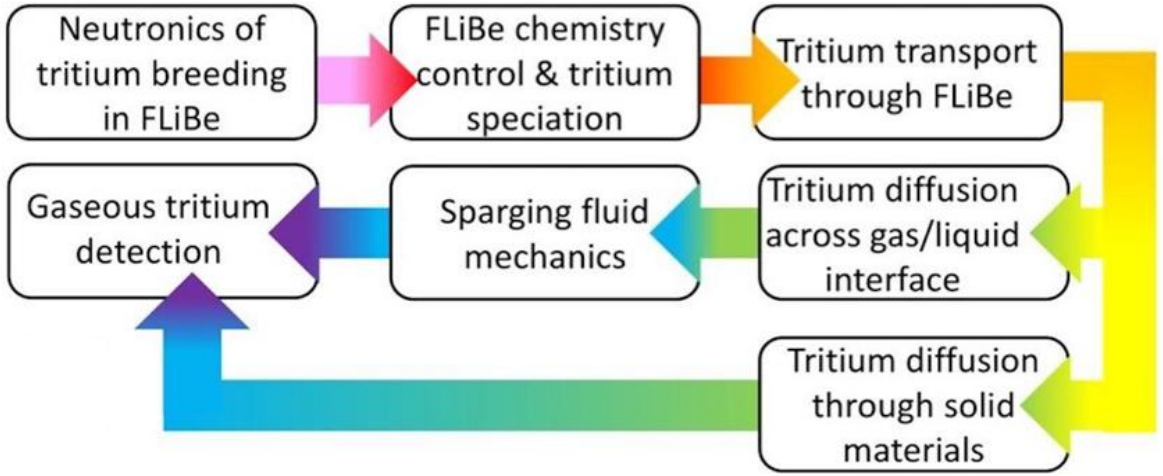
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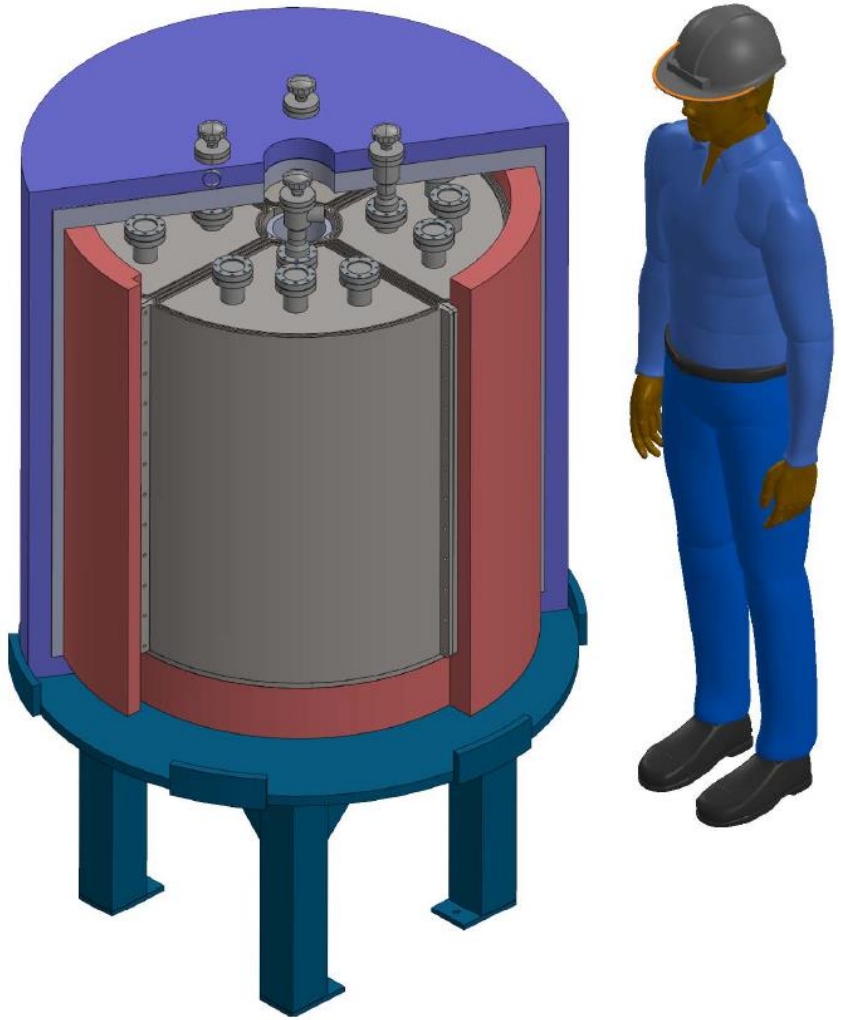




Modular tanks for parametric studies of materials, built-in repeatability, and future expansion

Liquid Immersion Blanket specific studies

LIBRA: Liquid Immersion Blanket Robust Accountancy



LIB tritium areas to evaluate:

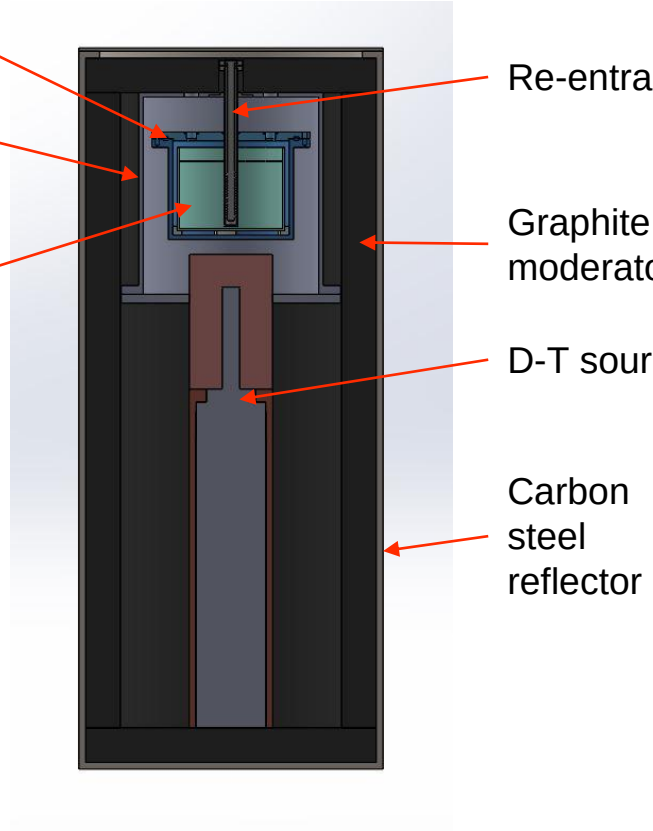
- **Breeding with D-T neutrons**
- **Transport in FLiBe**
- **Permeation**
- **Extraction**
- **Detection**
- **Global accounting**

LIB FLiBe areas to evaluate:

- **Chemical control**
- **Material compatibility**
- **Composition tolerances**
- **Fluid dynamics**

Smaller scale test this winter: first irradiation of FLiBe with 14 MeV D-T neutrons





- 1 L of FLiBe
- 14.1 MeV isotropic D-T neutron source with 3×10^8 n/s fluence (ThermoFisher)
- Tyne Engineering tritium detection system
- OpenMC for neutronics
- FISPACT-II for material activation

